202.29 Proof Rolling and Stabilizing Crushed Base Course – Before acceptance, each compacted course of base course aggregate shall receive one complete coverage by the tires of a truck having a 9 tonne single axle dual tire or 17 tonne tandem axle group with dual tires with a tire pressure of 600 kPa. The truck shall be loaded to its gross vehicle weight and driven at a speed of 1 to 2 m/s.

There shall be no deflection, reaction (noticeable movement in a localized area directly under the vehicle tire) or pumping of the ground surface (vertical displacement of top surface not directly under vehicle tires) observed under the moving vehicle's tires. This testing shall be performed under the observation of the Ministry Representative.

Any areas where rutting or displacement occurs shall be either excavated or replaced and proof rolled or stabilized by the addition of suitable blending material incorporated uniformly into the base to the satisfaction of the Ministry Representative.

The supply, load, haul, placing, proof rolling, and mixing of such stabilizing aggregates as necessary to correct deficiencies in aggregate stability shall be incidental to the Work for Base aggregates. Blending may be performed at the pit or quarry or on the highway in a manner acceptable to the Ministry Representative.

202.30 Benkelman Beam Testing – At any time during the course of the work, when considered necessary by the Ministry Representative, Benkelman Beam testing may be undertaken at the Ministry's expense.

MEASUREMENT AND PAYMENT

202.31 Water Applied to the Highway – Water for compaction or dust control shall be incidental to the bid price for the granular material.

202.32 Development and Cleanup of Ministry-Provided Pits and Quarries – Authorized clearing and grubbing of Ministry provided pits and quarries and construction of the access road is the responsibility of the Contractor. Removal and stockpiling of overburden/topsoil during development of the proposed work areas, and spreading overburden on a depleted pit or quarry shall be the responsibility of the Contractor and the work shall be completed to the satisfaction of the Ministry Representative.

Unless explicitly stated otherwise in the Special Provisions, no separate payment will be made for clearing, grubbing, disposal or relocation of stockpiles, debris or contaminated materials, or for any other costs of site preparation, pit development, remediation, or access, or for any delay or other cost arising from the use of Ministry-provided pits or quarries by the Contractor, and all costs thereof shall be covered in the prices for the Items under which payment is provided for the applicable materials.

202.33 Stockpile Reject Aggregate

202.33.01 Aggregate from screening or crushing operations in a Ministry-provided pit or quarry, whether:

(a) Oversize under SS 202.09, or

(b) Rejected at the crusher in order that the product meets the requirements of SS 202.05,

shall be stockpiled separately according to size in accordance with this SS 202. Stockpiles of reject aggregate shall not be contaminated by organic or other deleterious materials.

202.33.02 No Payment – No payment will be made for any reject aggregate.

202.34 Blending Materials for Stabilizing Aggregates on the Highway – Blending materials, supplied, loaded, hauled, placed and mixed into the granular aggregate on the highway to correct deficiencies in aggregate stability, will be paid for at the same rate as the granular aggregate which has to be stabilized. The final blended gradation must still meet the design aggregate gradation.

Normally, no additional payment will be made to cover the costs of work required to correct stability failure in the aggregate unless the Ministry Representative is satisfied that such instability is inherent in the type of aggregate available to the Contractor from a mandatory use Ministry-provided pit or quarry, and that the lack of stability has not arisen because of unsatisfactory production methods or improper construction practices. In these circumstances, the Ministry Representative will issue an Order for Extra Work to the Contractor to cover the additional cost of mixing blending material into the granular aggregate to correct the stability deficiency.

202.35 Aggregate Supplied in Stockpile Only – Aggregate, supplied in stockpile only, will be paid for at the Unit Price bid per CUBIC METRE, measured in stockpile and computed by surface to surface or, at the Ministry Representative's option, average end areas.

Payment at the unit price bid price shall be accepted as full compensation for everything furnished and done to supply (where applicable), produce and stockpile the crushed aggregate.

202.36 Aggregates Supplied in Place

202.36.01 Unit Price Payment – Crushed or screened aggregates (surfacing, base, subbase and bridge end fill) supplied and constructed in place will be paid on a neat line basis for at the applicable Unit Price bid per CUBIC METRE. Payment shall be full compensation for everything furnished and done, including without limitation costs for acquisition, development, and payment of royalties for private pits or quarries, purchase price of aggregate, access road maintenance, screening, crushing, stockpiling, loading, hauling, spreading, compaction and moisture adjustment (watering or drying) in place as specified.

202.36.02 Advance Payment: Crushed or Screened Aggregates Supplied in Place – An advanced payment of the greater of 70% of the demonstrated production costs or 30% of the extended amount for the Item will be made for crushed or screened aggregates (surfacing, base or subbase) in stockpile. No advance payment will be made for pit run aggregates. Recovery of the advance payment will be made on each progress estimate as the respective aggregate is subsequently withdrawn from the stockpile, with an additional adjustment on the final progress estimate for any material remaining in stockpile such that the entire advance payment is recovered.

202.36.03 Surplus Aggregate – Where actual aggregate needs on the Contract fall below the Approximate Quantity indicated on Schedule 7, the Ministry will compensate the Contractor at the rates indicated in Table 202-F for a portion of surplus aggregates produced by the Contractor which meet all the applicable quality and gradation specifications of the contract.

For each classification of aggregate, the maximum quantity of surplus aggregate eligible for compensation shall be the Approximate Quantity of that aggregate on Schedule 7 minus the quantity of that aggregate paid on the final progress estimate.

There will be no compensation for surplus aggregate for pit run aggregates or for any portion of the production on other aggregates which are beyond the Approximate Quantity. Where aggregates are purchased from a pre-existing commercial pit or quarry operation, there will be no compensation for any aggregate beyond that actually used in the Contract.

Surplus aggregates, originating from private land, in stockpile on Ministry lands are the property of the Ministry and no additional compensation will be paid. Surplus aggregate stockpiles on private lands will be the property of the Contractor and there will be no compensation for any aggregate beyond that actually used in the contract.

Table 202-F: Compensation for Surplus Aggregate

Stockpile Location	Compensation for Surplus Aggregates ($/m^3)	
	Crushed	Screened
Ministry land	$7.50	$3.75
Private land	$5.00	$2.50

202.36.04 Ministry Materials Processed on Private Lands – Where the Contractor elects to source raw materials from a Ministry pit or quarry, but process those materials on private lands:

(a) The Contractor shall extract only as much raw material from the Ministry source as is necessary to produce the quantity of product required to be incorporated in the final Project, inclusive of staging and maintenance requirements, with a small additional contingency amount determined by the Contractor and the Quality Manager and approved by the Ministry Representative.

(b) Ministry-sourced materials shall not be used in any way to improve the private lands, including without limitation for such purposes as access development, maintenance, or stockpile site preparation, unless approved in advance by the Ministry Representative under any conditions the Ministry Representative may require.

(c) Any surplus raw or processed material remaining on the private lands at the end of construction shall:

(i) be returned by the Contractor to the Ministry source and stockpiled at a location determined by the Ministry Representative, all at the Contractor's expense; or

(ii) upon request by the Contractor and at the Ministry Representative's sole discretion, the Contractor may be offered the options of:

(A) of leaving the materials on the private land and providing a written agreement with the owner of the property which shall state that the Ministry will have free access to and exclusive use of the remaining materials for a period of twenty-four (24) months after the completion of the Contract, after which time the Ministry relinquishes all claims to the material; or

(B) purchasing the remaining material at a negotiated amount to be back-charged to the Contractor.

(d) Where the volume of material remaining is less than the additional contingency amount determined pursuant to SS 202.36.04(a), the Ministry Representative will make the offers in SS 202.36.04(c)(ii). If agreement is not reached, SS 202.36.04(c)(i) will apply

202.37 Stockpile Volume Determination - The Ministry will determine the volume of aggregate by surveying the stockpile(s) using string-line techniques, determine volume using prismoidal volumes between surfaces, and deducting 10% for stockpile loss.

Alternatively, the Ministry Representative may elect to survey using cross section techniques and/or determine volumes using end-area volumes.

The base of the pile will be determined from a pre-stockpile survey or, where such a survey is not available, from a surface determined by the Ministry Representative as being

a reasonable interpolation from the intersection of the pile sides with the adjacent ground level.

Where there are two or more stockpiles of aggregate meeting the same gradation classification, the compensation will be based on the cumulative quantity of those stockpiles.

APPENDIX 202-A

FRACTURE COUNT FOR COARSE AGGREGATE (BCH 1-13)

202-A.01 Purpose – This test determines the amount of fractured material in the coarse part of an aggregate sample.

(a) **Method "A"** is used for crushed granular surfacing and base aggregates.

(b) **Method "B"** is used for paving aggregates.

202-A.02 Apparatus

(a) Drying oven, preferably forced draft, capable of maintaining a uniform temperature of 110 ± 5 º C.

(b) A nest of sieves of the following sizes:

- **Method "A"**: 37.5 mm, 25.0 mm, 19.0 mm, 12.5 mm, 9.5 mm, 4.75 mm
- **Method "B"**: 19.0 mm, 13.2 mm, 9.5 mm

(c) Balance with sufficient capacity and accurate within 0.1 percent of the mass of sample.

202-A.03 Test Sample – The sample of aggregate to be tested shall be representative, oven-dry, and reduced to an amount suitable for testing. The mass of the sample shall conform to Table 202-G.

Table 202-G: Test Sample Mass

Maximum Nominal Size (mm)	Minimum Dry Mass (kg)
37.5	4.0
25.0	2.5
12.5	1.5
9.5	1.0

202-A.04 Procedure

202-A.04.01 Method "A"

(a) Sieve the coarse aggregate to be tested on the 37.5 mm, 25.0 mm, 19.0 mm, 12.5 mm, 9.5 mm 4.75 mm sieves.

(b) Separate each sieve size into two groups, fractured and unfractured. Each rock in the fractured group shall have **at least one** fractured face or shall be naturally angular with sharp edges. (See Discussion 1).

(c) For each sieve size, count the number of fractured rocks and the total number of rocks (fractured and unfractured).

(d) Calculate and record the % fracture for each sieve size and report the total % fracture for the sample using Equation 202-A-1 and Equation 202-A-2.

202-A.04.02 Method "B"

(a) Separate the coarse aggregate to be tested from the fine aggregate by sieving it over the 4.75 mm screen.

(b) Separate the coarse aggregate for each sieve size into two groups, fractured and unfractured. Each rock in the fractured group shall have **two or more** fractured faces produced by a crusher or shall be naturally angular with sharp edges. Each rock in the unfractured group shall be unfractured or fractured only in one dimension.

(c) Obtain the mass of the fractured group, also the mass of the total coarse aggregate (fractured + unfractured groups).

(d) Calculate and report the total % fracture for the sample using Equation 202-A-3.

202-A.05 Discussion

(a) Fractured face shall be defined as a fracture in any plane whose area is 15% or more of the largest cross section area in a parallel plane.

(b) The required sample can be conveniently obtained from the separated fractions of the Dry or Wash Sieve Test.

Equation 202-A-1: Percent Fracture for Each Sieve Size (e.g. 19 mm)

$$\%\ \text{Fracture (19 mm)} = \frac{\text{Number of Fractured Rocks (19 mm)}}{\text{Total No. of Rocks (19 mm)}}\ \text{x } 100\%$$

Equation 202-A-2: Percent Fracture for the Total Sample (by Number of Rocks)

$$Total\ \%\ Fracture = \frac{Number\ of\ Fractured\ Rocks\ (All\ Sieve\ Sizes)}{Total\ No.\ of\ Rocks\ (All\ Sizes)}\ x\ 100\%$$

Equation 202-A-3: Percent Fracture for the Total Sample (by Mass)

$$\%\ Fracture = \frac{Mass\ of\ Fractured\ Rock}{Mass\ of\ Total\ Rocks}\ x\ 100\%$$

APPENDIX 202-B

PETROGRAPHIC ANALYSIS TEST

202-B.01 Purpose – The purpose of the Petrographic Analysis Test is to identify the various rock types and rock characteristics in the aggregate fraction of material retained on the 9.5 mm sieve of a gravel deposit.

The test is performed after a Micro Deval soundness test or a magnesium sulphate soundness test indicates that the granular materials are weak, non-durable, or suspect. The petrographic analysis test is to prove or determine the cause of the poor performance of the aggregate and to determine the extent or contributing factors of specific rock types or the extent of deleterious materials or clay particles. This information will be used to determine the possible use of the materials.

202-B.02 Scope – The test shall be performed by a qualified professional with experience in rock type identification, rock classification and gravel/aggregate sources and use; and shall be presented in a signed report that details findings. Recognized procedures for rock identification shall be used.

The procedures involve simple physical and chemical tests, such as the hardness scratch test, hydrochloric acid reaction and visual surface examination assisted by magnifying lens. The predominance of easily distinguishable types in petrographic samples will permit reasonably accurate testing. Important accessories are a sample kit, the rock and mineral descriptions and a good text.

Rock types are to be listed by percentage weight under the broad categories of origin (sedimentary-metamorphic-igneous) with condition descriptions (i.e. hardness, weathering and cementation). Note that the descriptions of quality regarding durability or soundness (good, fair, poor, deleterious) are vital to the test results.

Additional information on general condition including weathering, shape, porosity, coatings etc. shall also be recorded.

202-B.03 Apparatus

(a) Balance – 20,000 gram capacity, accuracy ± 0.1 gram.

(b) Oven capable of maintaining 110°C.

(c) Sieves 19 mm (3/4 inch) and 9.5 mm (3/8 inch) and shaker.

(d) Sample splitter with pans.

(e) Hand-lens (10X magnification).

(f) A ten percent solution of hydrochloric acid.

(g) Aluminum pans.

(h) A stereoscopic microscope.

202-B.04 Sample Preparation – A 1500 gram sample of representative aggregate passing 19 mm and retained on 9.5 mm screen is required.

(a) For pit run materials, a minimum of 18 kg, or one full gravel bag of representative aggregate is required.

(b) This material is oven-dried at 110°C ± 5°C overnight, or for 16 hours.

(c) After drying, the sample is quartered in the sample splitter.

(d) A one-quarter sample is sieved to obtain passing 19 mm aggregate. Additional quarters are sieved if the initial quarter provides less than 1500 grams of the required sizes.

(e) The passing 19 mm retained on 9.5 mm aggregate is mixed and 1500 grams separated by splitter.

(f) Fractured materials are separated from non-fractured materials and a fracture count test shall be performed using Method A (APPENDIX 202-A).

(g) Following removal (by hand) of clay balls and very soft weak material, the non-fractured rocks are washed to assist visual identification of mineral characteristics. The soft fractured rocks shall be washed with great care as their degradation may prevent correct identification. The clay balls and lumps shall be weighed and the percent of sample noted.

202-B.05 Rock Type Classification – Assignment of each particle to rock type and character is required.

Identification of specific rock type may be preceded by general identification under the major rock categories of: igneous, sedimentary and metamorphic. Following simple physical, chemical and visual tests, each particle is then identified with a specific rock type and description.

Rock types are weighed to the nearest 0.1 gram. Weights are recorded on the test sheet and the percentage of each is calculated from the total sample weight.

Comments on each rock type include descriptions of physical and chemical qualities such as:

(a) Hardness: hard, medium, soft.

(b) Weathering: non weathered, slightly or intensely weathered.

(c) Porosity

(d) Cementation: firmly cemented to friable.

(e) Coatings

202-B.06 Quality Characterization (good, fair, poor, deleterious)

202-B.06.01 Basis of Quality System – The following quality distinctions are relative estimates of a rock's physical and chemical condition and of probable engineering quality.

- **Good:** Particles are hard, durable, free from fracture potential, little or no capillary absorption.
- **Fair:** Particles are soft but sound and tough, medium hard, slightly to moderately weathered, have small to moderate capillary absorption, are relatively smooth and impermeable.
- **Poor:** Particles are soft to very soft, friable, potential slaking when wetted and dried, highly fractured potential, capillary absorption moderate to high.
- **Deleterious:** Particles react chemically with alkali in Portland Cement causing expansion in concrete, although they may be suitable in base course.

202-B.06.02 Report Submission Requirements – The report shall describe the tests performed, present the test results, including descriptions of weathering, shape, porosity, coatings, angularity and provide a petrographic number.

A description of suitability shall be stated by the individual performing the test.

The report shall contain an executive summary of rock types and conditions similar to that shown in the example below.

"e.g. Igneous granodiorite 90% hard, non-weathered. volcanics 10% hard and fresh, with some volcanics (30% of volcanics) slightly weathered and some volcanics (5% of volcanics) are vesicular, soft and intensely weathered. No clay lumps, some particles (less than 5% of sample) contained coatings of clay."

202-B.06.03 Sample Petrographic Number – The Petrographic Number (PN) for the sample is calculated by multiplying the percentage of each of the four quality classifications by the following multipliers:

- Good 1
- Fair 3
- Poor 6
- Deleterious 10

and then summing the results and determining the Overall Sample Rating in accordance with Table 202-H.

Table 202-H: Petrographic Number and Overall Rating

Sample Petrographic Number	Overall Sample Rating
100 to 125	Good
>125 to 140	Fair
>140 to 155	Poor
> 155	Deleterious

SECTION 204
ROCK CUTS

DESCRIPTION

204.01 Scope – This Section describes the controlled blasting techniques required to produce rock cut slopes at the locations shown on the Drawings or as authorized by the Ministry Representative. Rock shall be excavated to subgrade elevation as shown on the Drawings. The work shall be conducted, whether by blasting or other methods, in a manner that complies with SS 165, minimizes blast damage to the excavation backslope, and minimizes the requirement for stabilization. The responsibility for the results of every blast lies with the Contractor. Nothing within this specification shall be interpreted to mean the Ministry accepts responsibility for the results of any blast.

204.02 Definitions

(a) **Backline Holes** – A line of holes detonated or intermittently detonated along the specified excavation backslope.

(b) **Bench –** A near-horizontal or shallow slope surface at the top of a near-vertical rock face where blast holes are collared.

(c) **Bench Width –** The horizontal distance on the bench between the excavation backslope and the crest of the existing rock face. Generally, this dimension defines the width of the rock to be removed and is perpendicular to the highway alignment.

(d) **Blast Crater** – A local depression in rock formed by blasting.

(e) **Buffer Holes** – The line of production holes located closest to and parallel to the backline holes.

(f) **Burden** – The distance between the blast hole and the nearest free face.

(g) **Charge** – A specific quantity of explosive such as a cartridge.

(h) **Controlled Blasting** – Controlled blasting is the controlled use of explosives and accessories in carefully spaced and aligned production and backline drill holes to produce the specified excavation backslope within the specified excavation limits. The purpose of controlled blasting is to minimize damage to the rock backslope, to help ensure long-term stability and to minimize flyrock.

(i) **Controlled Delay Sequence** – The delay sequence required for controlled blasting.

(j) **Cushion Blasting** – A blasting method whereby the production holes are detonated before the backline holes.

(k) **Delay** – A blast initiator with a specific delay time period before detonation.

(l) **Excavation Backslope or Smooth Wall** – This is the location of the free surface or shear plane in the rock produced by blasting along the backline holes.

(m) **Excavation Limits** – This is the boundary within which rock is removed.

(n) **Explosive Decoupling –** The separation of an explosive charge from the wall of the blast hole.

(o) **Flyrock** – The throw of fragmented material during blasting.

(p) **Free Face** – The rock face that provides relief for a blast.

(q) **Guide Holes** – Those holes along the backline that are not loaded with explosive and are not stemmed.

(r) **Hole Deviation –** Borehole misalignment distance measured in two directions, within and perpendicular to the excavation backslope. Within the plane of the excavation backslope, hole deviation is borehole misalignment as measured by the horizontal offset distance between any point in the design hole location and the point at a corresponding depth in the actual hole. Perpendicular to the plane of the excavation backslope, hole deviation is the horizontal offset distance between the plane of the design excavation backslope and the centreline of the borehole being measured.

(s) **Lift** – The vertical distance between the top and bottom of an area to be blasted.

(t) **Overbreak in Rock** – Overbreak in rock is any rock outside of the specified excavation limits (except slide material described in SS 201.42) that is excavated, displaced or loose due to the inherent character of any formation encountered or due to any other cause.

(u) **Overburden** – This is Type D material (as described in SS 201.11) overlying rock.

(v) **Presplit (or Pre-Shear) Blasting** – A blasting method whereby backline holes are detonated before production holes are detonated.

(w) **Production Blasting** – Blasting of production holes detonated in a controlled delay sequence.

(x) **Production Holes** – All holes within the specified excavation limits that are not backline holes. Production holes are often more widely spaced than backline holes.

(y) **Rock Excavation** – Rock excavation includes drilling holes, loading holes with explosives, detonation by

controlled blasting, mucking to subgrade and ditch level, and removal of material to a designated area. Rock excavation also includes secondary breakage of oversize rock.

(z) Rock – Rock is Type A material as defined in SS 201.11.01.

(aa) Scaling – The removal of all loose material using scaling bars, portable hydraulic jacks, other hand tools, wire rope cables, compressed air blow pipes, blasting without the need for drilled holes and other methods authorized by the Ministry Representative.

(bb) Setback Distance – The setback distance is the horizontal distance between the top of the excavation backslope and the toe of the overburden material. Setback is created by removing overburden.

(cc) Spacing – The distance between blast holes perpendicular to the burden.

(dd) Stemming – Material placed on top of the explosive to provide confinement of the explosive gases.

(ee) Subdrill – The distance drilled below subgrade level.

(ff) Trimming – The removal by drilling and controlled blasting of potentially unstable rock remaining at the exposed excavation backslope. Trimming may not require backline holes.

204.03 Quality Control

204.03.01 General – Quality control shall be conducted for all blasting operations.

204.03.02 Blasting Consultant – The Contractor shall retain a blasting consultant specialist, acceptable to the Ministry Representative, to provide quality control. The consultant shall not be an employee of the Contractor, explosives manufacturer or explosives distributor.

204.03.03 Vibration Specialist – The Contractor shall provide a blast vibration monitor if required. The blast vibration monitoring shall be under the direction of the Vibration Specialist.

204.03.04 Quality Control Activities For Blasting Operations – Quality control for blasting operations shall be performed for test sections, full scale blasting operations and trimming. A field report generated by the Consultant shall be provided to the Ministry Representative within one day of each site visit by the Consultant. Quality control for blasting includes, but is not limited to the following activities by the blasting consultant:

(a) Viewing the proposed blast area at least one (1) week before drilling operations commence.

(b) Preparing and submitting blast designs for production and backline holes.

(c) Intermittently viewing the site during drilling for each blast area.

(d) Intermittently observing the loading of holes with explosives and tying in to the delay system.

(e) Observing the blasts and reviewing the excavated areas.

(f) Attending on site to view the excavation backslope after each lift and reviewing excavation backslope for quality control purposes.

(g) Viewing the site while reviewing proposed changes to the blast design and preparing new designs.

(h) Attending on site at other times as are appropriate to assist the Contractor in the setting up, assessment and adjustment of the various procedures to be employed for blasting.

(i) Preparing a field report for each site visit, including details of the progress of blasting operations, a statement whether the blast design is being complied with, any changes to the blast design, any recommendations made to the Contractor and any problems encountered by the Contractor.

204.04 Submittals

204.04.01 General – The Ministry may perform Quality Audits. The receipt of submittals by the Ministry shall not relieve responsibility from the Contractor for the accuracy and adequacy of the submittals. Submittals are for quality assurance and record keeping purposes. Inadequate or incorrect submittals will be returned for revisions prior to acceptance. The Contractor shall submit to the Ministry Representative the following documentation for review.

204.04.02 Qualifications and Experience of Contractor – The Contractor shall provide a statement of the qualifications, experience and work function of all personnel assigned to drilling and blasting duties. A statement of previous work experience on similar projects shall also be provided. This statement shall include the project name, location, volume of rock, year constructed and the owner/client contact name. The company, the driller and the blaster shall each have a minimum of 5 consecutive years demonstrated experience in drilling and controlled blasting work on at least 3 projects involving rock cuts over 8 m height along transportation corridors.

204.04.03 Qualifications and Experience of Proposed Blasting Consultant – The Contractor shall provide the name of the consulting firm, and the name and qualifications of the blasting consultant's on-site representative who will be providing the quality control for rock excavation. The blasting consultant shall have a minimum of 5 consecutive years demonstrated experience in preparing successful blast designs along transportation corridors for at least 3 projects. The following information shall be included in the qualifications submitted:

(a) Project name, location and experience.

(b) Name and phone number of owner/client contact who can verify the experience of the blasting consultant's site representative.

204.04.04 Qualifications and Experience of Proposed Vibration Specialist – The Contractor shall provide the name and qualifications of the vibration specialist who will be providing the quality control for rock excavation. The vibration specialist shall have a minimum of 5 consecutive years demonstrated experience in the field of vibration monitoring for at least 3 projects. The following information shall be included in the qualifications submitted:

(a) Project name, location and experience.

(b) Name and phone number of owner/client contact who can verify the experience of the vibration specialist.

204.04.05 Proposed General Construction Plan – The Contractor shall submit a general construction plan to the Ministry Representative showing proposed blasting sequence numbers defining the order of blasts for the contract. The plan shall also show the following information for each blast:

(a) Blast sequence number.

(b) The location of the blast in relation to the remaining rock on the site.

(c) Approximate dimension of the rock to be removed.

(d) Estimated volume of rock to be removed.

(e) Location of the disposal site.

204.04.06 Sequence of Operations – The Contractor shall adopt a logical, systematic sequence of operations to ensure blasting is conducted safely and effectively. The following general sequence of blasting operations outlines minimum requirements of the Contractor to maximize the stability of the excavation backslope and does not limit or supersede any other requirements in this specification:

(a) Retain a blasting consultant to perform quality control.

(b) Submit blast designs for review and authorization by the Ministry.

(c) Accurately survey locations of proposed blast holes.

(d) Drill holes, load explosives, detonate blast and remove muck pile.

(e) Check the excavation backslope to determine hole offsets, hole angles, hole alignment and compliance with allowable tolerances.

(f) Review blast results and make changes to blasting operations as necessary.

(g) Perform backslope stabilization to the satisfaction of the Ministry Representative before subsequent lifts are detonated.

204.04.07 Blast Design – The Contractor shall provide and follow a blast design, approved and signed by the blasting consultant, not less than one week prior to commencing drilling and blasting operations and a minimum of one day before the Contractor proposes to implement any changes to the previously utilized drilling or blasting methods. The design may be prepared by the blaster, but shall be reviewed by the Contractor and forwarded to the blasting consultant for signing. The design shall contain full details of the drilling and blasting patterns and controls that the Contractor proposes to use for controlled blasting. The blast design shall contain the following minimum information:

(a) Date the design was prepared and proposed date of blast.

(b) Station limits of proposed blast.

(c) Plan and section views of proposed drill pattern including free face, burden, blasthole spacing, blast hole diameters, blast hole angles, lift height, drill offset, hole depth, and subdrill depth.

(d) The location of production (including buffer) and backline holes.

(e) Loading diagram showing type and amount of explosives, primers, initiators, and location and depth of stemming.

(f) Initiation sequence of blast holes including delay times and delay system.

(g) Manufacturer's data sheets for all explosives, primers, delays, and initiators to be used.

(h) Blasting consultant's signature, printed name, and company name.

(i) Blaster's signature, printed name, company name, and blaster's certificate number.

204.04.08 Pre-blast Survey – The Contractor shall conduct a pre-blast survey a minimum of one day before blasting operations commence. The pre-blast survey shall include a complete description of the existing condition of any nearby buildings, structures, wells and utilities that potentially may be damaged by blasting operations. The survey method used shall be acceptable to the Contractor's insurance company.

204.04.09 As-Built Blast Design – Within one day after each blast, the Contractor shall submit, unless otherwise authorized by the Ministry Representative, an as-built blast design showing all actual blast details in a format that permits direct comparison with the proposed blast design.

204.04.10 Blasting Consultant's Field Report – The Contractor shall provide the blasting consultant's field report to the Ministry Representative within one day after each visit by the blasting consultant.

204.04.11 Vibration Control Records – As required, the Contractor shall provide all seismograph records of

vibration monitoring and interpretation of results within one day after each blast to the Ministry Representative.

204.04.12 Peak Overpressure Records – As required, the Contractor shall provide a permanent signed and dated record of the peak overpressure measurements within one day after each blast to the Ministry Representative.

MATERIALS

204.11 Explosives and Related Products

204.11.01 Manufacturer – All products and materials used for any blasting operations shall be products of a company regularly engaged in the manufacture of explosives and related products.

204.11.02 Expired Explosives – Explosives with an expired shelf life shall not be used.

204.11.03 Water Resistance – Water resistant explosives may be required for the work.

CONSTRUCTION

204.31 Permits and Regulations – The Contractor shall obtain all necessary permits and shall comply fully with the laws, rules and regulations of Municipal, Provincial and Federal agencies in connection with the use, transportation, storage and safe handling of all explosives, including those regulations contained in the WCB Occupational Health and Safety Regulation.

204.32 Supervision – The Contractor shall provide at least one person thoroughly trained and experienced in the use of explosives who shall be present at all times during the execution of all blasting operations and who shall direct such work.

204.33 Personnel – The Contractor shall ensure that all persons conducting blasting operations have a valid blaster's certificate issued by the WCB or is under the direct supervision of a certificate holder.

204.34 Safety – The Contractor shall meet all WCB regulations. All work shall be performed in a manner that prevents injury or harm to any personnel employed in the rock excavation area. Warning signs shall be posted and readily recognizable audible warning signals shall be used. The perimeter of the area affected by blasting operations shall be patrolled and controlled by direct voice communication. The Ministry Representative shall stop the work if the safety of the public is being jeopardized by the Contractor's blasting operations.

204.35 Flyrock Control – Before the detonation of any blast in areas where flying rock or other debris may result in personal injury or damage to property, the area within the excavation limits shall be covered with suitable blasting mats, soil or other equally serviceable material to prevent flyrock.

204.36 Other Damage – The Contractor shall be responsible for any damage resulting from blasting. Occupants of local buildings shall be notified by the Contractor, prior to the commencement of the blasting, as to the timing, size of blasts, types of warning and other signals.

204.37 Subgrade and Ditch Areas

204.37.01 General – The subgrade shall be constructed to ±50 mm of the specified line and grade before the placement of Select Granular Subbase Material. Any pinnacles of intact rock protruding above the design subgrade elevation shall be removed.

204.37.02 Supporting Rock – The Contractor shall exercise care and use appropriate methods to prevent breaking, loosening or otherwise damaging supporting rock below subgrade level and ditch bottom. The Contractor shall be responsible for the methods used and for any damage to the rock structure resulting from the operations.

204.37.03 Drainage – Excavated rock shall be free draining unless otherwise authorized by the Ministry Representative. If craters formed in rock by blasting below subgrade elevation are not free draining, then the Contractor shall provide drainage by trenching to a free outlet. These blast craters and drainage trenches shall be backfilled to subgrade elevation.

204.37.04 Shatter Zone for Drainage – Unless otherwise authorized by the Ministry Representative, drainage shall also be provided by shattering the upper portion of rock below subgrade and ditches within the entire excavation limits. The shatter zone shall be 300 mm thick. A shatter zone thickness of 900 mm may be required near cut to fill transition zones for 10 m beyond the transition, or as requested by the Ministry Representative.

204.37.05 Backfill Material – Backfill shall be free draining material such as clean broken rock or coarse clean granular material from a source authorized by the Ministry Representative. All backfill shall be placed and compacted as specified.

204.37.06 Ditches – The ditches shall be formed and cleaned before any base material is placed on the subgrade.

204.38 Blasting Test Section(s)

204.38.01 General – All requirements for full scale blasting shall also apply to test sections unless otherwise authorized by the Ministry Representative. Prior to commencing full-scale blasting operations, the Ministry Representative may require the Contractor to demonstrate the adequacy of the proposed blast design by drilling, blasting, and excavating short test sections, up to 30 m in length, to determine which combination of method, hole spacing, timing, and charge

yields acceptable results. The length of the blast test section shall be as specified in the blast design.

204.38.02 Reduced Section Length – Where requested by the Ministry Representative, the Contractor shall use test section lengths of less than 30 m.

204.38.03 Backline Hole Spacing – The Contractor shall begin the tests by drilling backline holes 750 mm apart along the specified excavation backslope, then adjust if needed, until the Ministry Representative authorizes the spacing to be used for full-scale blasting operations.

204.38.04 Review of Test Section Results – The Contractor shall not drill beyond the test section until it has been excavated and the results reviewed by the blasting consultant and continuation authorized by the Ministry Representative.

204.38.05 Revision of Methods – If either the blasting consultant or the Ministry Representative determines that the results of the test section are unsatisfactory, then notwithstanding the Ministry Representative's prior review of such methods, the Contractor shall adopt such revised methods as are necessary to achieve the required results.

204.38.06 Additional Test Sections – If at any time during the progress of the work the methods of drilling and blasting do not produce the required excavation backslope geometry within the tolerances specified for backline holes, then the Contractor will be required to drill, blast and excavate short sections, not exceeding 30 m in length, until a technique is achieved that will produce the desired results.

204.39 Overburden Removal and Setback – The setback distance shall be 3.0 m. Unless otherwise authorized by the Ministry Representative, the Contractor shall, before drilling the backline holes, remove all overburden within the excavation limits or 10 m beyond the limits of the production holes in a direction parallel to the backline. The overburden surface shall be sloped at 1.5:1 unless otherwise authorized by the Ministry Representative.

204.40 Backline and Production Holes

204.40.01 Stemming – The upper portion of all holes between the topmost charge and the hole collar shall be stemmed. Stemming materials shall be sand or other inert angular granular material with similar specific gravity, passing a 9.5 mm sieve.

204.40.02 Hole Obstructions – Before placing charges, the Contractor shall determine that the hole is free of obstructions for its entire depth. All necessary precautions shall be exercised so that the placing of the charges will not cause spalling of material from the walls of the holes.

204.40.03 General Orientation – All holes shall be drilled downward unless otherwise specified in the blast design and authorized by the Ministry Representative. In general, slash holes (horizontal, near horizontal or fanned out holes) shall not be drilled along the excavation backslope or on pioneering routes excavated to provide access for backline hole drilling, unless authorized by the Ministry Representative.

204.40.04 Insertion of Explosive – Explosive materials shall not be inserted into the holes until the blast design has been reviewed by the Ministry Representative.

204.41 Backline Holes

204.41.01 General – The Contractor shall control the hole layout and drilling operations using proper equipment and techniques to ensure that backline holes are located and oriented correctly. The proposed location of each backline hole shall be accurately surveyed and staked.

204.41.02 Hole Location – For each blast, the line of backline holes shall extend 10 m to 20 m beyond the limits of the production holes to be detonated or to the end of the specified excavation backslope, as applicable.

204.41.03 Hole Location Tolerance – Backline holes shall be drilled within 75 mm of the staked collar location.

204.41.04 Hole Deviation – Backline holes shall not deviate from the plane of the specified excavation backslope by more than 150 mm as measured perpendicular to the slope. Backline holes shall not deviate more than 150 mm as measured within the plane of the excavation backslope.

204.41.05 Hole Diameter – The **backline** holes shall be 50 mm to 75 mm in diameter, or as recommended by the blasting consultant and authorized by the Ministry Representative. Trim blast holes may be as small as 25 mm diameter.

204.41.06 Hole Length – The length of backline holes for any individual lift shall not exceed 8 m unless the Contractor can demonstrate to the Ministry Representative that the Contractor can stay within the tolerances and produce the required excavation backslope geometry.

204.41.07 Guide Holes – Guide Holes shall be of the same diameter and drilled in the same **plane** and to the same tolerance as the remaining backline holes.

204.41.08 Hole Length Increase – Upon satisfactory demonstration, the length of holes may be increased to a maximum of 12 m with written authorization of the Ministry Representative. If more than 5% of the backline holes are misaligned in any one lift, then the Contractor shall reduce the height of the lifts until the 150 mm alignment tolerance is met.

204.41.09 Control of Drilling Orientation – All drilling equipment used to drill the backline holes shall have mechanical devices attached to that equipment to accurately determine the orientation of the drill steel entering the rock. Backline hole drilling will not be permitted if these devices are either missing or inoperative.

204.41.10 Offset Between Lifts – When the cut height requires more than one lift, a maximum 0.5 m offset

between lifts is permitted to allow for drill equipment clearances. The Contractor shall begin the backline hole drilling on the deign backline for the top lift, and shall adjust the start of lower lifts to compensate for drill offset and any drift which may have occurred in the upper lifts. The excavation angle required to compensate for drill offset can be determined as shown in APPENDIX 204-C, Figure 204-1.

204.41.11 Length for Toe Berm Removal – Drilling 0.5 m below ditch bottom is permitted to facilitate removal of the toe berm.

204.42 Presplit Blasting

204.42.01 General – Unless otherwise specified in the blast design and authorized by the Ministry Representative, presplit blasting shall be conducted for all blasting to the excavation backslope. In general, cushion blasting may proceed when the bench width is less than three times the lift height.

204.42.02 Explosive Type and Accessories – Drill hole conditions may vary from dry to filled with water. The Contractor shall use explosives and blasting accessories appropriate for the drill hole conditions encountered to accomplish the specified results. Only standard explosives manufactured for presplit blasting shall be used in backline holes, unless otherwise specified in the blast design and authorized by the Ministry Representative. Bulk ammonium nitrate and fuel oil (ANFO) shall not be loaded into the backline holes.

204.42.03 Explosive Decoupling – Explosives shall be evenly distributed and decoupled to the maximum extent possible.

204.42.04 Explosive Charges – The bottom charge of backline holes may be larger than the remaining charges but shall not be large enough to cause overbreak. The top charge of backline holes shall be placed far enough below the collar and be sufficiently small to avoid overbreak and heaving of rock beyond the excavation backslope.

204.42.05 Modified Blasting Sequence – The Contractor may detonate the backline holes before drilling production holes, provided satisfactory excavation backslopes are obtained.

204.42.06 Backline Hole Delays – If required to reduce ground vibrations or noise, backline holes may be delayed, provided the effective hole-to-hole delay time is not more than 25 ms.

204.42.06 Excavation Backslope Geometry – The excavation backslope shall not deviate more than 150 mm from a plane passing through adjacent drill holes except where the character of the rock is such that, as determined by the Ministry Representative, irregularities are unavoidable. The 150 mm tolerance shall be measured perpendicular to the plane of the slope. Should any blast result in a non-compliant backwall, the blast design shall be modified, prior to any subsequent blasting, to prevent future non-conformance.

In no case shall any portion of the slope encroach on the ditch.

204.43 Cushion Blasting

204.43.01 General – Where the horizontal distance from the specified excavation backslope to the existing free face is less than 5 m, the Contractor may use cushion blasting instead of presplit blasting.

204.43.02 Delay Time – The difference in delay time between the backline holes detonated after the buffer holes shall be between 25 ms and 75 ms. With the exception of these criteria, requirements for presplit blasting shall also apply to cushion blasting.

204.44 204.44 Production Blasting

204.44.01 Hole Diameter – Production holes shall not exceed 150 mm in diameter, unless otherwise specified in the blast design and authorized by the Ministry Representative.

204.44.02 Hole Location – Production blast holes (not including buffer holes) shall not be drilled closer than 2.0 m to the backline holes, unless otherwise specified in the blast design and authorized by the Ministry Representative. The bottom of the production holes (including buffer holes) shall not be lower than the bottom of the backline holes.

204.44.03 Delay Sequence – All sequences shall be delayed in such a manner that successive delays promote the movement of rock in the direction of a free face at all times.

204.44.04 Damage to Backslope – It is the Contractor's responsibility to take all necessary precautions during production blasting to minimize blast damage to the excavation backslope.

204.44.05 Buffer Holes – A line of buffer holes shall be drilled along a plane parallel to the backline holes. Buffer hole diameters shall be between 50 mm and 75 mm unless otherwise specified in the blast design. Unless otherwise specified in the blast design, the line of buffer holes shall be drilled approximately 1 m away from the backline holes and spaced 1.0 to 1.5 m centre to centre.

204.44.06 Buffer Hole Charge and Detonation – The explosive charge in buffer holes shall not exceed 50% of the full explosive load that could be placed in a 75 mm diameter production hole. Detonation of the buffer holes shall be on a delay sequence toward a free face. Ammonium nitrate and fuel oil shall not be used for buffer holes unless otherwise specified in the blast design and authorized by the Ministry Representative.

204.45 Vibration Control and Monitoring

204.45.01 Vibration Control – The Contractor shall use blasting methods designed to limit the intensity of ground

vibrations originating within the excavation limits. When blasting near buildings, structures, wells, utilities or other works that may be subject to damage from blast induced ground vibrations, the ground vibrations shall be controlled using properly designed delay sequences and allowable charge weights per delay.

204.45.02 Charge Weight Per Delay – Allowable charge weights per delay shall be based on vibration levels that will not cause damage. The Ministry Representative may monitor vibration levels at the blast site by requesting trial blasts to determine actual vibration levels reached during blasting.

204.45.03 Monitoring – Monitoring shall meet the International Society of Rock Mechanics (ISRM) standards. Whenever vibration damage to adjacent structures is possible, the Contractor shall monitor each blast with approved seismograph(s) located between the blast area and the closest structure(s) subject to potential blast damage. The geophone shall be placed as close as possible to the structure(s) but not directly above the structure(s). The seismograph(s) shall be set to record particle velocity, accelerations, and frequency in the range generally found with controlled blasting. The peak particle velocity shall be calculated as the maximum vector sum of three mutually perpendicular components of vibration. All components and peak particle velocity shall be recorded.

204.45.04 Vibration Limits – Peak particle velocity, accelerations and frequency shall not be allowed to exceed the safe limits of the nearest structure subject to potential vibration damage. The Contractor shall employ a qualified vibration specialist to establish the safe vibration limits.

204.45.05 Interpretation – The vibration specialist shall interpret the seismograph records to ensure that the seismograph data is effectively used in the control of the blasting operations with respect to the existing structures.

204.46 Air Concussion and Noise Control

204.46.01 General – When requested by the Ministry Representative, an air concussion monitoring system shall be installed in a representative location between the blasting area and the nearest structure subject to potential blast damage or annoyance. The equipment used to make the air concussion measurements shall be the type specifically manufactured for that purpose.

204.46.02 Monitoring – The air concussion monitoring system shall be set to record air overpressure on the linear setting. When requested by the Ministry Representative, human annoyance (A weight setting) and human ear response (C weight setting) shall also be recorded.

204.46.03 Overpressure – Peak overpressure shall be controlled using appropriate blast hole patterns, detonation systems and stemming to prevent venting of blasts, and to minimize air concussion and noise levels produced by the blasting operations. The contractor shall use a qualified vibration specialist to establish safe overpressure limits.

204.47 204.47 Excavation Backslope Stabilization

204.47.01 Stabilization – The excavation backslope shall be stabilized as recommended by the Ministry's geotechnical engineer and to the Ministry Representative's satisfaction, during or upon completion of the excavation of each lift. Unless otherwise authorized by the Ministry Representative, drilling of the next lift shall not proceed until stabilization has been completed. Drilling of the next lift shall not proceed until all concerns about stability raised by the Ministry's geotechnical engineer and the Ministry Representative are addressed by the stabilization work. Stabilization shall be completed before any base material is placed upon the subgrade. Stabilization shall be at the Contractor's expense.

204.47.02 Stabilization Methods – Stabilization methods include scaling, trimming, the application of rock bolts, shotcrete, slope mesh, drains or other stabilization techniques recommended by the Ministry's geotechnical engineer and authorized by the Ministry Representative.

204.48 Special Use of Excavated Rock – Excavated rock of suitable quality required for walls, riprap, paving, or other special use shall be sorted and deposited in stockpiles if so requested by the Ministry Representative.

MEASUREMENT

204.81 Type A Excavation – Type A excavation will be measured by the CUBIC METRE.

204.82 Excavation Backslope (Smooth Wall) – The excavation backslope will be measured by the SQUARE METRE only where the height of the cut exceeds 2 m, and shall be measured over the entire height of such cuts. Measurement will apply to the condition of the slope prior to scaling. Measurement for payment, including measurement of smooth wall surface area, hole deviation and slope deviation, will be conducted by the Ministry.

PAYMENT

204.91 Type A Excavation – Payment for TYPE A EXCAVATION will be at the Contract Unit Price per cubic metre and will be authorized by the Ministry Representative after submittal of all documentation required in this Section. The Unit Price will be considered full compensation for the requirements of this Section.

Design slope lines including approved offsets, subgrade and ditch lines shall represent the pay lines of Type A excavation.

If so determined by the Ministry Representative, stabilization necessitated by improper blasting operations shall be at the Contractor's expense.

Trimming will be paid at the Contract Unit Price per cubic metre for TYPE A EXCAVATION.

All costs associated with blasting test sections and any revised methods necessary to produce acceptable results shall be incidental to Type A excavation.

Additional drilling, blasting and excavation required to construct subgrade to the required tolerances and excavating, loading, hauling, placing and compacting backfill material for blast craters and drainage trenches to subgrade level shall be incidental to Type A excavation.

All costs associated with vibration control, air concussion and noise control, and all monitoring shall be considered incidental to Type A excavation.

204.92 Excavation Backslope (Smooth Wall) – Payment for EXCAVATION BACKSLOPE will be at the Contract Unit Price per square metre for only that part of the exposed cut face meeting the geometric tolerances for excavation backslope in this Section. Geometric tolerances to be met include hole location tolerance, hole deviation, offset between lifts and deviation of excavation backslope geometry.

Scaling of loose material disturbed by blasting including removal and disposal of overbreak shall be considered incidental to payment for excavation backslope.

Where there is no Excavation Backslope (Smooth wall) Item in Schedule 7, the Excavation Backslope (Smooth wall) Work shall be performed incidental to Type A Excavation.

APPENDIX 204-A

PROPOSED BLAST DESIGN

File #: _____
PROJECT NO. _____

(TO BE SUBMITTED NOT LESS THAN 1 DAY BEFORE DRILLING)

GENERAL:

DATE PREPARED_____

CONTRACTOR'S NAME _____ PROPOSED BLAST DATE _____
BLASTER'S NAME _____ BLAST # _____
BLASTER'S CERTIFICATE NUMBER _____

HIGHWAY#_____
BLAST LOCATION AT: _____

UTILITY PROTECTION OR STANDBY REQUIRED: (CIRCLE) YES NO

SITE DETAILS:

HEIGHT ABOVE HWY _____ (m) SLOPE ANGLE _____ (°) SLOPE LENGTH _____ (m)
ROCK TYPE _____
DISTANCE TO NEAREST STRUCTURE (UTILITY)_____ (m)

ANTICIPATED DRILLING DETAILS:

PROPOSED NO. OF BACKLINE HOLES_____ PROPOSED APPROX. NO. OF HOLES _____
PROPOSED AVG. DEPTH _____ (m) HOLE DIA. _____ (mm)
PROPOSED NO. OF ROWS _____ APPROX. HOLE SPACING _____ (m) BURDEN _____ (m)
PROPOSED MAXIMUM DEPTH _____ (m) PROPOSED HOLE INCL.: (CIRCLE) VERTICAL HORIZ. VARIABLE
PROPOSED TOTAL DEPTH _____ (m)

PROPOSED BLAST DETAILS:

EXPLOSIVE TYPE _____ EXPLOSIVE SIZE _____mm by _____mm
APPROX. NUMBER OF CARTRIDGES _____ APPROX. TOTAL WEIGHT _____ (kg)
PROPOSED NUMBER OF DELAYS _____ DELAY TYPE AND LENGTH _____ (ms)

INITIATION DEVICE: (CIRCLE) E.B. CAPS SAFETY FUSE OTHER (SPECIFY) _____
BLASTING MACHINE: TYPE _____ CAPACITY _____

APPROX. VOLUME OF ROCK TO BE BLASTED _____ (m^3)

PROPOSED BLAST SKETCH:

SHOW THE FOLLOWING:	PLAN VIEW:	CROSS SECTION VIEW:
SCALE PLAN VIEW CROSS SECTION ROCK GEOMETRY APPROX. HOLE LOCATIONS APPROX. HOLE DEPTH PROPOSED TIE-IN PATTERN ROW BY ROW DELAYS DETONATION DIRECTION HIGHWAY LOCATION UTILITY LOCATION NORTH ARROW		

BLASTING CONSULTANT NAME _____ SIGNATURE_____ COMPANY_____

BLASTER'S SIGNATURE _____ COMPANY_____

RECEIVED BY MINISTRY REPRESENTATIVE (SIGNATURE) _____

(Use Additional Sheets if Necessary)

APPENDIX 204-B

AS-BUILT BLASTING RECORD

File #: _____
PROJECT NO. _____

(TO BE SUBMITTED NOT MORE THAN 1 DAY AFTER EACH BLAST)

GENERAL:

DATE PREPARED_____

CONTRACTOR'S NAME _____ ACTUAL BLAST DATE _____
BLASTER'S NAME _____ BLAST # _____
BLASTER'S CERTIFICATE NUMBER _____

HIGHWAY#_____
BLAST LOCATION AT: _____

UTILITY PROTECTION OR STANDBY USED: (CIRCLE) YES NO TYPE OF UTILITY PROTECTION _____

SITE DETAILS:

HEIGHT ABOVE HWY _____ (m) SLOPE ANGLE _____ (°) SLOPE LENGTH _____ (m)
ROCK TYPE _____
DISTANCE TO NEAREST STRUCTURE (UTILITY)_____ (m)

DRILLING DETAILS:

NUMBER OF BACKLINE HOLES_____ TOTAL NUMBER OF HOLES _____
AVERAGE DEPTH _____ (m) HOLE DIAMETER _____ (mm)
NUMBER OF ROWS _____ HOLE SPACING _____ (m) BURDEN _____ (m)
MAXIMUM DEPTH _____ (m) HOLE INCLINATION: (CIRCLE) VERTICAL HORIZ. VARIABLE
TOTAL DEPTH _____ (m)

BLAST DETAILS:

EXPLOSIVE TYPE _____ EXPLOSIVE SIZE _____mm by _____mm
TOTAL NUMBER OF CARTRIDGES _____ TOTAL WEIGHT _____ (kg)
TOTAL NUMBER OF DELAYS _____ DELAY TYPE AND LENGTH _____ (ms)

INITIATION DEVICE: (CIRCLE) E.B. CAPS SAFETY FUSE OTHER (SPECIFY) _____
BLASTING MACHINE: TYPE _____ CAPACITY _____

VOLUME OF ROCK BLASTED _____ (m^3)

BLAST SKETCH:

SHOW THE FOLLOWING:	PLAN VIEW:	CROSS SECTION VIEW:
SCALE PLAN VIEW CROSS SECTION ROCK GEOMETRY HOLE LOCATIONS HOLE DEPTH TIE-IN PATTERN ROW BY ROW DELAYS DETONATION DIRECTION HIGHWAY LOCATION UTILITY LOCATION NORTH ARROW		

BLASTER'S SIGNATURE _____ COMPANY_____

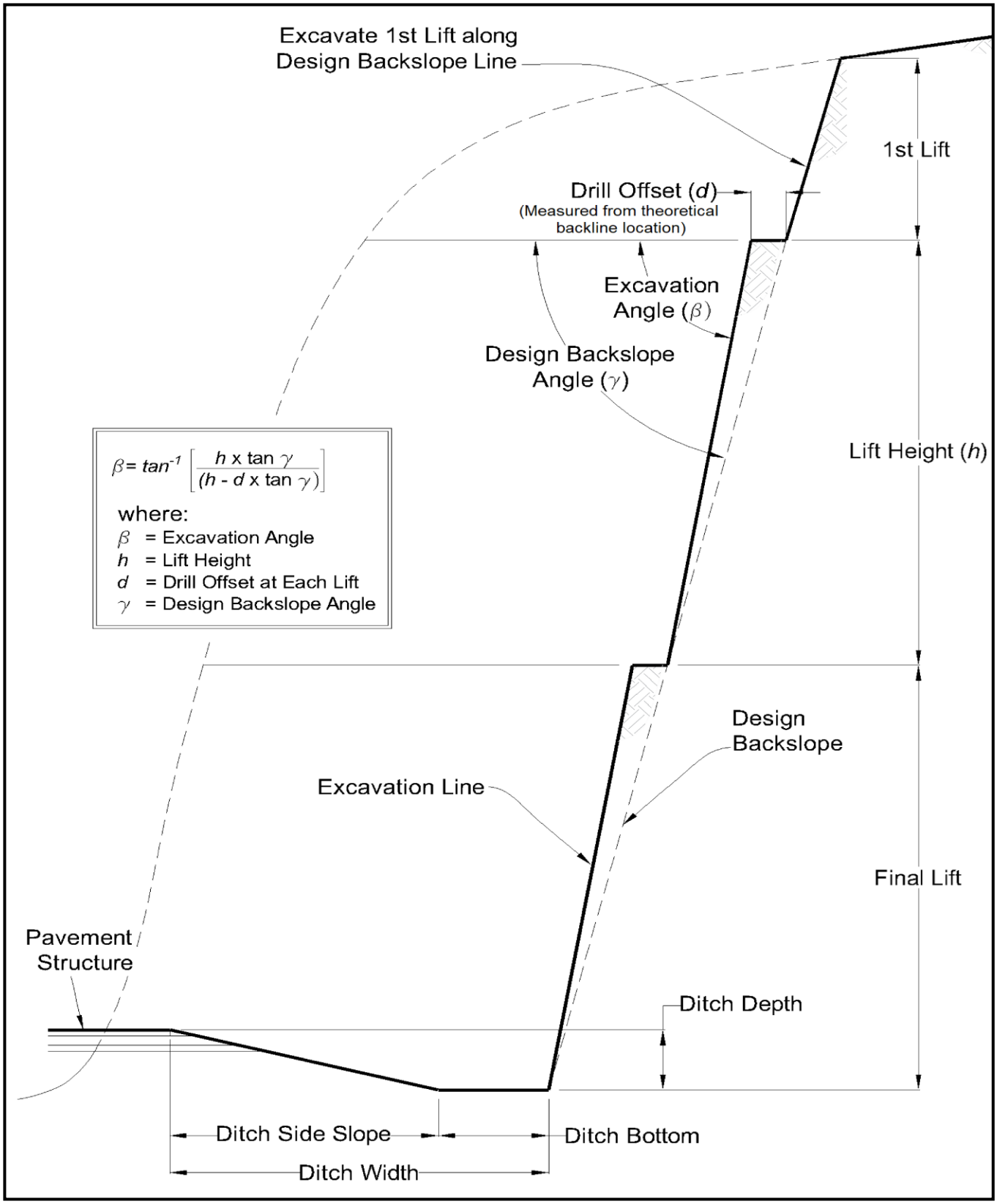
RECEIVED BY MINISTRY REPRESENTATIVE (SIGNATURE) _____

(Use Additional Sheets if Necessary)

APPENDIX 204-C

CALCULATION OF EXCAVATION ANGLE FOR DRILL OFFSET

Figure 204-1: Calculation of Excavation Angle *(β)* for Drill Offset



SECTION 205

RIPRAP

205.01 General – This Section covers the protection of embankments and channels, by using the Class of riprap, at the designated locations, specified in the Drawings and Special Provisions.

Work within any watercourse shall be carried out in conformity with the environmental protection provisions in the Contract and SS 165, all to the satisfaction of the Ministry Representative.

205.02 Quality Control

205.02.01 General – The Contractor shall be fully responsible for all quality control testing, inspection and documentation to achieve compliance with the Contract.

205.02.02 Material Testing

(a) Initial Testing and Gradation – The Contractor shall provide documented test results for the riprap material properties, per Table 205-A and the gradation of the riprap to the Ministry Representative for review at least one week prior to starting riprap delivery to the Site.

(b) Gradation Sampling Areas – For riprap gradation, quality control shall continue to occur throughout the project and include, but not be limited to the following:

(i) gradation of representative samples of riprap at the source while production is underway; and

(ii) gradation of the placed riprap at the Site while placement is underway and at completion of the placement.

(c) Gradation Sampling Frequency – Gradation quality control shall occur at a frequency of not less than once per week, or once per 1000 tonnes or 500 m^3 of riprap produced or placed, whichever is more frequent, at each source and placement site that is active at any time during the week.

Table 205-A: Tests for Riprap Material Properties

Property	ASTM Test Designation	Allowable Value
Specific Gravity	D6473	≥2.50
Absorption	D6473	≤2%
Soundness by use of Magnesium Sulphate	D5240	≤10% (following 5 cycles)
Micro–Deval Abrasion Loss Factor	D6928	≤20%

205.02.03 Gradation Control – The Contractor may elect to perform gradation quality control using either mass or dimensional assessment in accordance with the following.

(a) Light Riprap – For riprap of Class 50 or less, quality control may be done visually, supplemented with measuring dimensions or masses as necessary to confirm the accuracy of the visual assessment.

(b) Heavy Riprap by Mass – The quality control for the gradation of the Class 100 and heavier riprap by mass shall:

(i) Conform to the gradation specified in using Table 205-B; and

(ii) be done in accordance with the ASTM D5519 Method A or Method B, or other method approved in advance by the Ministry Representative.

(c) Heavy Riprap by Dimension – The quality control for the gradation of Class 100 and larger riprap by size shall:

(i) Conform to the gradation specified in Table 205-C with a tolerance of -5% to +15 % of the values listed; and

(ii) be done using ASTM D5519 Method C or Method D, the FHWA FLH T 521 Wolman count method, or other method approved in advance by the Ministry Representative.

(d) Documentation – Documentation of the gradation testing and inspection shall be provided to the Ministry Representative within 24 hours from its completion.

205.02.04 Hold Points – The following hold points, as defined in SS 145.12, will apply for Class 100 kg riprap and larger unless otherwise directed by the Ministry Representative:

(a) Gradation QC and material property test results for riprap are provided to the Ministry Representative prior to starting delivery to Site

(b) Completion of Toe/Terminal end key excavations

(c) Preparation of back slope/surface

(d) Application of filter(s)

(e) Completion of toe construction

205.03 Quality Assurance – The Ministry may carry out quality assurance by auditing the Contractor's quality control program and by testing and inspection at its discretion. Ministry quality testing shall not relieve the Contractor of responsibility for providing quality control.

205.04 Riprap Properties

205.04.01 General - Rock shall be hard, durable, and angular quarry rock of a quality that will not disintegrate on exposure to water or the atmosphere.

205.04.02 Size and Gradation – The size and gradation of rocks shall be in accordance with the following:

(a) well-graded;

(b) Table 205-B for mass or Table 205-C for dimensions; and

(c) the minimum dimension of each individual rock shall be greater than one-third of its maximum dimension.

205.04.03 Material Properties – Rocks used for riprap shall only break with difficulty, have no earthy odour, no closely spaced discontinuities, and should not absorb water easily. Rocks composed of appreciable amounts of clay or silt shall not be accepted for use as riprap.

Any riprap source shall be tested by the Contractor for conformance to the requirements of Table 205-A, prior to use on the Site.

Where the Ministry has provided a riprap source and has previously tested its material properties, the Contract will indicate that, and the Contractor will not be required to undertake further material property testing.

Where the Contractor elects to use another source, or the Ministry does not have test results for an available Ministry source, the Contractor will be responsible to test to, and ensure compliance with, the allowable values for the tests given in Table 205-A.

Representative samples may be broken off representative riprap rocks and crushed to allow performance of the tests.

Rocks shall be tested for Acid Rock Drainage and Metal Leaching as required by the Ministry Representative. Testing shall meet the Ministry requirements outlined in the Ministry's Technical Circular T–04/13, available at the link below.

https://www2.gov.bc.ca/assets/gov/driving–and–transportation/transportation–infrastructure/engineering–standards–and–guidelines/technical–circulars/2013/t04–13.pdf

205.05 Preparation – Areas to receive riprap shall be trimmed to a uniform surface and to the slope(s) indicated on the Drawings or as directed by the Ministry Representative.

Before rock placement commences, loose material shall be removed and minor hollows filled with surrounding native materials well tamped–in to the approval of the Ministry Representative.

Table 205-B: Gradation of Rock by Class of Riprap

Class of Riprap (kg)	Rock Mass (kg)			
	Percentage Smaller Than Given Rock Mass			Max. Size
	15%	50%	85%	
10	1	10	30	50
25	2.5	25	75	125
50	5	50	150	250
100	10	100	300	500
250	25	250	750	1 250
500	50	500	1 500	2 500
1000	100	1 000	3 000	5 000
2000	200	2 000	6 000	10 000
4000	400	4 000	12 000	20 000

Table 205-C: Gradation and Intermediate Dimension of Rock by Class of Riprap

Class of Riprap (kg)	Intermediate Dimension (mm)			
	Percentage Smaller Than Intermediate Dimension			Max. Size
	15%	50%	85%	
10	90	200	285	350
25	125	270	385	450
50	155	340	485	600
100	200	425	610	750
250	270	575	830	1 000
500	340	725	1 050	1 250
1000	425	915	1 325	1 600
2000	535	1 150	1 650	2 000
4000	675	1 450	2 100	2 500

Note: Table 205-C shows the intermediate dimension as defined in the Wolman method as per FHWA FLH T 521 corresponding to the rock mass shown in Table 205-B, based on spherical volume, using Specific Gravity = 2.50. Regardless of actual source Specific Gravity, the dimensions indicated remain applicable (subject to the limits specified in Table 205-A).

205.06 Filter Layers – Filter layers and placement, where required, shall be as per the Drawings and Special Provisions or as directed by the Ministry Representative.

205.07 Foundations and Placement – The nominal thickness and the surface width for each Class of riprap shall conform with the requirements specified in Table 205-D.

Each truckload of rock brought to the site shall provide a complete range of the rock sizes in the gradation.

Work shall be carried out to prevent cracking or breaking of rock riprap by crushing under machine tracks. Work shall be carried out to avoid disturbing the filter layer(s). Damage shall be repaired at the Contractor's expense.

Placement shall be as indicated on the Drawings or as directed by the Ministry Representative.

Layout and installation details for riprap at bridge embankments and for embankment protection works parallel to waterway flow shall be in accordance with SS Drawing SP205-1, unless otherwise specified. SS Drawing SP205-1 shall not apply for riprap for culvert end treatment or around in-stream piers.

At the toe of sloped riprap, a sufficient number of the larger rocks shall be placed to form a firm foundation. The remaining larger rocks shall be distributed evenly throughout the mass.

Rocks shall be placed to the required thickness, providing a reasonably well–graded mass with the minimum of voids. Clusters of small or large stones shall be avoided.

205.08 Machine Placed Riprap – The controlled placement of rock of the Class specified shall produce a well–graded rock mass of the nominal or required thickness over the area indicated. Placement of riprap shall not be by end–dumping. The rock shall be machine manipulated as necessary to provide mass stability and a regular surface with a minimum of voids.

205.09 Hand–Laid Riprap – Hand–laid riprap, normally Class 10 or 25, shall conform to the size, gradation and requirements set out in SS 205.04. Individual rocks too large to handle shall be machine manipulated for satisfactory setting and spacing.

205.10 Grouted Riprap – Where grouted riprap is shown or required, the surfaces of the rocks shall be cleaned and wetted and the interstices filled with cement mortar, well rodded and pounded in for a minimum mortar depth of 300 mm or as otherwise detailed or required by the Ministry Representative. The mortar shall consist of one-part Portland cement to three parts well–graded clean fine aggregate (1:3) mixed to a proper consistency.

Table 205-D: Placement Dimensions by Class of Riprap

Class of Riprap (kg)	Nominal Thickness of Riprap* (mm)	Surface Width, W* (mm)	
		2H:1V Slope	1.5H:1V Slope
10	350	783	631
25	450	1006	811
50	550	1230	992
100	700	1566	1262
250	1000	2236	1803
500	1200	2684	2163
1000	1500	3355	2704
2000	2000	4473	3606
4000	2500	5591	4507

* See SS Drawing SP205-1 for the description of the Nominal Thickness and Surface Width dimension "W".

MEASUREMENT

205.11 Measurement – Measurements will be made by multiplying the facial area by the average thickness dimensions as shown on the Drawings or as directed by the Ministry Representative. No allowance will be made for the quantity of rock placed in excess of these dimensions.

PAYMENT

205.12 Payment – Payment shall be made at the Unit Price bid per cubic metre for the Class of riprap specified or required. The Unit Price bid shall be accepted as full compensation for everything completely furnished and done in connection therewith, including supply, haul, placing, and quality control.

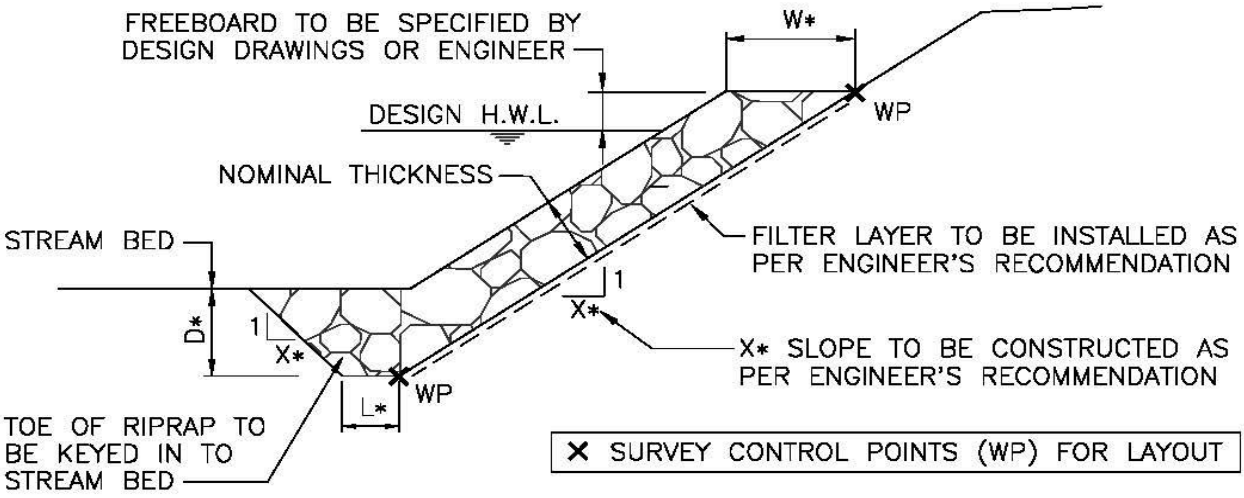
Payment for the excavation work at the placement site shall be paid for under "Roadway Drainage and Excavation" or "Foundation Excavation", whichever is specified in the Contract.

Where the source of supply for the riprap is off–Site or outside the design excavation limits as shown on the Drawings, the Unit Price for Riprap shall also include all costs to develop the source and produce the riprap.

Where the source of supply is on–Site and within the design excavation limits as shown on the Drawings, or within any slip, payment to excavate and haul the rock for riprap will be included within the Contract Unit Price for Type A, as defined in SS 201.

TYPICAL RIPRAP INSTALLATION AT BRIDGE EMBANKMENTS AND EMBANKMENT PROTECTION PARALLEL TO WATER FLOW

SP205–1



L* & D* TO BE SPECIFIED BY ENGINEER
W* TO BE AS PER TABLE 205–D
OR AS PER DESIGN DRAWINGS

NOTE: DETAIL NOT APPLICABLE TO CULVERT END PROTECTION

NOT TO SCALE
ALL DIMENSIONS ARE IN MILLIMETERS UNLESS OTHERWISE NOTED

BRITISH COLUMBIA | Ministry of Transportation & Infrastructure

SECTION 206
ROCK BOLTS

DESCRIPTION

206.01 General – Unless otherwise specified, the Post Tensioning Institute (PTI) D35.1-14 Recommendations for Prestressed Rock and Soil Anchors shall apply to rock bolts and other rock anchoring systems. Rock bolting operations shall include, but not be limited to, pre-construction and construction work including quality control, site preparation, selection of appropriate installation method, drilling, sealing, bar insertion, grouting and testing of rock bolts.

Rock bolts shall be installed in the areas designated by the Ministry Representative. Unless otherwise stated below, all rock bolts shall be installed and tensioned to the rock bolt manufacturer's specifications.

206.02 Codes and Standards

CSA G30.18	Carbon Steel Bars for Concrete Reinforcement
CSA G40.21	Structural Quality Steels
CSA A23.2-9C	Compressive Strength of Cylindrical Concrete Specimens.
ASTM A123	Standard Specification for Zinc (Hot-Dip Galvanized) Coatings on Iron and Steel Products
ASTM A153	Standard Specification for Zinc Coating (Hot-Dip) on Iron and Steel Hardware

206.03 Quality Control – The Contractor shall provide quality control for rock bolt operations.

206.04 Submittals – The Contractor shall supply the following to the Ministry Representative:

206.04.01 Proposed Rock Bolt System – Provide documentation as follows for the proposed rock bolt system for review at least 5 working days before commencing rock bolt installation.

(a) Rock Bolts – Type, dimensions and manufacturer of the bar, centralizers, face plate, washer (where applicable), and nut.

(b) Resin – Published specifications showing name of material, manufacturer, graph of unconfined compressive strength development versus time, gel time, viscosity, shelf life, storage and handling requirements.

(c) Grout – Published specifications showing name of material, manufacturer, 7 day and 28 day unconfined compressive strength, composition, shelf life, storage and handling requirements.

(d) Data Sheets – Upon request by the Ministry Representative, supply the manufacturer's product data sheets.

(e) Properties of Steel – Upon request by the Ministry Representative, supply the physical and chemical properties for each lot or heat number of the steel.

(f) Test Results – Upon request by the Ministry Representative, submit a certified statement from an independent testing laboratory of the physical dimensions and mechanical properties of the rock bolt bar proposed for use. This statement shall include:

- **(i)** Minimum cross sectional area of the threaded portion of the bar.
- **(ii)** Minimum cross sectional area of the unthreaded portion of the bar.
- **(iii)** Yield strength of the bar.
- **(iv)** Ultimate tensile strength of the bar.
- **(v)** Percent elongation of the bar at yield and ultimate strength.

(g) Quick Setting Mortar – Name of material, manufacturer, 7 day and 28 day unconfined compressive strength, and composition.

(h) Sealing – Proposed method of sealing fractures and voids in drill holes to prevent loss of grout.

206.04.02 Calibration Certificates – Provide tensioning jack calibration certificates seven (7) days before commencing rock bolt installation. Calibration of jacks shall have been performed by an authorized testing agency not more than 30 days prior to rock bolt testing. The certificate shall show the relationship between gauge pressure and applied load. Pumps and jacks shall be paired for calibration.

206.04.03 Contractor's Rock Bolt Records – Provide daily rock bolt tensioning records, in a form similar to that shown in APPENDIX 206-A, within one day of each day of rock bolt operations. Records shall include contractor name, date of tensioning, weather, temperature, test jack identification number, pump identification number, name of person who tested the rock bolt, rock bolt location, rock bolt number, rock bolt length, resin or grout details, test start time, test end time, gauge reading for each minute of the creep test, and lock off load. For resin systems, records shall also include the start and end times of spinning the bar through the resin in the hole.

206.05 Definitions – The following definitions refer to words and terms used in this Section. For definitions not covered

here, refer to definitions provided by the Post Tensioning Institute.

(a) Bonding Agent – Material such as grout or resin used to bond the rock bolt bar to the rock in the bond zone or free stressing length.

(b) Bond Zone – The distal or lower portion of the hole where the bonding agent transmits the applied tensile load to the surrounding rock.

(c) Free Stressing Length – The proximal or upper portion of the hole not within the bond zone.

(d) Grout Sock – A geotextile encasement within all or part of the drill hole, used to control grout loss in highly permeable rock conditions.

(e) Sealing Grout – Portland cement based grout that is pumped or gravity fed into the hole or used as drilling fluid to reduce the permeability of the rock immediately surrounding the hole.

MATERIALS

206.11 Materials – Steel materials shall be hot dip galvanized to ASTM A123 or ASTM A153, as appropriate. All resin, grout and galvanized steel materials shall be the products of established manufacturers regularly engaged in the manufacture of rock bolt materials for at least five years.

Materials shall meet the following additional requirements:

(a) Rock Bolt Bars

- **(i)** Steel hot-rolled Grade 517 MPa meeting CSA G30.18.
- **(ii)** Nominal solid bar diameter 22 mm unless otherwise specified.
- **(iii)** Threadlike surface deformations for full length of bar and suitable for mechanical coupling.
- **(iv)** Cut-thread reinforcing bar not permitted.

(b) Miscellaneous Hardware

- **(i)** Steel hardware to be compatible in size and strength with rock bolt bars.
- **(ii)** Face plates to be square and suitable for dome shaped nuts.
- **(iii)** Face plate size to be matched to bar size.
- **(iv)** Face plates to meet CSA G40.21 Grade 300W.
- **(v)** Face plate dimensions 10 mm by 150 mm by 150 mm unless otherwise specified.
- **(vi)** Face plates date stamped after galvanizing on the side visible when installed with the current year (in the format YYYY) in numbers 10 mm high.
- **(vii)** Face plates slotted for grout tube if grout is used.

(c) Resin

- **(i)** Fast-set and slow-set resin in cartridge form.
- **(ii)** A minimum unconfined compressive strength when fully mixed and cured of 90 MPa, tested in accordance with CSA A23.2.
- **(iii)** Encased in a plastic film that provides optimum resistance to moisture, and is easily ruptured to enable complete mixing during installation.
- **(iv)** Suitable thixotropic and viscous properties to permit adequate mixing of the resin components by rotation of the rock bolt bar and to contain the resin within the drill hole.
- **(v)** Easily identifiable gel time and as recommended by the resin manufacturer.
- **(vi)** Reach 80% of its ultimate strength within a time interval equal to five times the gel time.
- **(vii)** Non-shrink after the gel time.
- **(viii)** Unaffected by mild acids or mild alkalis.
- **(ix)** Cartridge boxes labelled with the resin expiry date.

(d) Grout

- **(i)** Pre-mixed, unsanded, non-metallic, and non-shrink cementitious grout containing silica fume.
- **(ii)** Can be mixed to a flowable consistency, typically with a water to cement ratio of 0.35 or as specified by the grout manufacturer to achieve or exceed the required strength.
- **(iii)** Minimum 7 day compressive strength of 30 MPa and a minimum 28 day compressive strength of 40 MPa, tested in accordance with CSA A23.2.
- **(iv)** Admixtures to be used according to the manufacturer's specifications and as authorized by the Ministry Representative.
- **(v)** Calcium chloride accelerator is not permitted.

(e) Mortar Pads

- **(i)** Portland cement based.
- **(ii)** Quick setting.

CONSTRUCTION

206.31 Execution

206.31.01 General – The entire rock bolt system shall be stored under cover away from deleterious materials. All grease and other deleterious material shall be removed from the steel prior to rock bolt installation.

206.31.02 Site Preparation – Where rock bolts may be adversely impacted, rock removal above and around proposed rock bolt locations shall be completed before installation of

rock bolts commences. Any minor rock scaling performed in conjunction with rock bolting shall be considered incidental to rock bolting.

206.31.03 Drill Holes

(a) Location, Orientation and Depth – The location, direction, angle and depth of the holes will be dependent on field conditions encountered and will be detailed by the Ministry Representative prior to and if necessary during construction.

(b) Hole Diameter – The diameter of the holes shall be suitable for the rock bolt system chosen. Where grout is used, the hole size shall be according to the rock bolt manufacturer's recommendations. Where resin is used, the hole size shall be according to the resin manufacturer's recommendations.

206.31.04 Cleaning – All water, grease, oil, cuttings and other deleterious materials shall be removed from finished holes by a water and/or air jet as required.

206.31.05 Sealing – The Contractor shall take measures incidental to rock bolting to maintain open drill holes, seal holes and prevent anchor grout loss as follows:

(a) Fill fractures and small voids;

(b) Bridge large voids;

(c) Consolidate around the hole.

(d) including but not limited to using the following measures:

(e) Sealing grouting;

(f) Lining the hole with steel or pvc casing;

(g) Grouting and re-drilling;

(h) Using grout sock to contain anchor grout

Alternative sealing methods shall be as authorized by the Ministry Representative.

206.31.06 Installation

(a) Installation Methods

(i) Rock bolt bars shall be surrounded and secured in the hole using one of the following two methods (See SS Drawing SP206-01):

(A) Grout in bond zone, grout in free stressing length.

(B) Resin in bond zone, grout in free stressing length.

(ii) A third method may be used as follows, if authorized by the Ministry Representative, due to field conditions:

(A) Resin in bond zone, resin in free stressing length.

(b) General – Rock bolts shall be inserted (or rotated) into the drill holes and fully encapsulated in resin or grout to the drill hole collar. When resin is used, the bolt shall be advanced and rotated at a rate recommended by the resin manufacturer.

(c) Anchorage Length – The anchorage length or bond zone shall be the last 1000 mm of the inserted end of the bar, unless otherwise specified.

(d) Centralizers – If grout is used, centralizers on 3.0 m centres shall centralize the rock bolt in the drill hole before grout is placed. Centralizers shall be suitable for holes in rock and shall be specifically manufactured for rock bolt application.

(e) Resin – Resin cartridges shall be installed as follows or as specified by the resin manufacturer:

(i) Fast Set Resin – A sufficient number of fast-setting cartridges shall be placed in the bond zone at the bottom of the hole for the anchorage.

(ii) Slow Set Resin – A sufficient number of slow-setting cartridges shall be placed in the free stressing length between the bond zone and the collar of the hole.

(f) Grout – Grout shall be prepared and placed as follows unless otherwise specified by the grout manufacturer:

(i) Mixing – Grout shall be mixed in a colloidal or high shear grout mixer according to the grout manufacturer's published instructions. Mixing paddles shall be slotted and perforated. Mixing time shall be not less than two minutes.

(ii) Batching – All ingredients for the grout mix shall be batched by mass. Water shall be added to the drum first and dry ingredients afterwards. Grout shall not be re-tempered after initial mixing. Grout shall be placed immediately after mixing.

(iii) Grout Placement – Grout shall be pumped or gravity fed through a grout tube extending to the bottom of the hole. The inserted end of the tube shall remain below the level of the grout in the hole to affect a continuous air free column as the grout level rises. Grout shall be placed quickly and continuously to avoid overworking, segregation, bleeding and disturbance of initial set. Grout that has stiffened due to delay in placing shall not be used in the work and shall be disposed of at an authorized location.

(g) Rock Face Preparation – The bearing surface shall be prepared to allow the face plate to be oriented within the limits recommended by the anchor manufacturer. If necessary, rock shall be chipped from around the face plate contact area.

(h) Mortar Pad Construction – Mortar pads shall be constructed as required to ensure the bar alignment is

within twenty degrees (20°) of a line perpendicular to the face plate. Mortar pads shall be installed where face plates deform excessively under load. The pad shall not crack or deform when loaded. Sufficient time shall be provided to allow pads to achieve sufficient bearing capacity prior to test-tensioning.

(i) End Hardware Installation

(i) Nuts shall bear uniformly against the face plate.

(ii) The bolt extension beyond the nut shall be 100 mm ± 10 mm.

206.31.07 Tensioning – All rock bolts shall be test-tensioned and locked off following set-up (or curing) of the bond zone and before set-up of the free stressing length. Prior to testing, the bond zone grout and/or resin shall meet the strength specified by the anchor manufacturer. The following procedure applies to 22 mm diameter Grade 517 MPa steel bars. An alternative procedure may be specified for different bar sizes.

(a) Equipment – Equipment required for tensioning shall be supplied by the Contractor and shall be of a size adequate to provide the required tension. A torque wrench shall not be used for tensioning.

(b) Test-Tensioning and Creep Test – Rock bolts shall be test-tensioned to 186 kN (42,000 lb). The test load shall be held for 10 minutes for the creep test.

(c) Lock-Off Tension – Rock bolts shall be locked-off to a design tension of 140 kN (31,500 lb) after testing.

(d) Acceptance Criteria – During the creep test a load loss of greater than 10% of the load applied shall be indicative of anchorage failure. Creep movement at the anchor head shall not exceed 2 mm during the creep test. A replacement rock bolt shall be installed at the Contractor's expense where these criteria are not met.

206.31.08 Rock Bolt Evaluation – The Ministry Representative will implement a program of evaluation of rock bolts installed. The Contractor shall perform the following test procedure on randomly selected bolts. After locking off the anchor, the load shall be re-applied to determine the lift-off load. The lift-off load shall be the tension level at which the anchor nut can be loosened by hand. Lift-off tests shall be performed on rock bolts chosen by the Ministry Representative to a minimum of 5% of the total number of rock bolts. One additional lift-off test on a different bolt shall be performed for each bolt whose lift-off load is not within 10% of the specified lock-off load. Following lift-off testing, all bolts shall be locked off as specified.

MEASUREMENT

206.81 Rock Bolts – Rock bolts will be measured by the METRE installed. The measurement length shall be the length of bar in the rock.

PAYMENT

206.91 Rock Bolts – Payment for ROCK BOLTS will be at the Contract Unit Price per metre. Payment for rock bolts will be authorized after installation to the contract specifications and after submittal of the completed Contractor's Daily Rock Bolt Testing and Tensioning Record. The Unit Price will be considered full compensation for all work and materials supplied according to the requirements of this Section.